Is the Band Worth It? A Multi-Seed Comparison of RMSNorm Variants

An empirical evaluation of BandRMS, ZeroCenteredRMSNorm, and ZeroCenteredBandRMSNorm across five regimes

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Abstract

We empirically evaluate three increasingly elaborate variants of RMSNorm normalization — BandRMS (output magnitude constrained to a learnable band $[1-\alpha, 1]$), ZeroCenteredRMSNorm (mean subtracted before RMS rescaling), and ZeroCenteredBandRMSNorm (both modifications combined) — against vanilla RMSNorm across five settings: ViT-pico on CIFAR-10, ResNet-18 on CIFAR-100, a 4-layer pre-norm transformer on IMDb, a 24-block deep residual stack under mixed precision, and a layer-level synthetic microbenchmark. We run n=5 random seeds per cell and report mean $\pm$ std, 95% bootstrap CI, and a paired sign-flip permutation p-value (B=10,000, Phipson-Smyth corrected) against the RMSNorm baseline. Four mechanistic probe callbacks log per-epoch activation mean, per-sample RMS distribution, weight L_2 trajectory, and gradient norm at every normalization layer, directly testing each variant's specific theoretical claim. Three findings emerge. First, **the band constraint by itself produces no measurable benefit on any task we tested**: BandRMS mechanistically confines output RMS to $[0.9, 1.0]$ as advertised, but yields zero accuracy or loss improvement over RMSNorm across all five experiments and ten parameter-matched cells. Second, **zero-centering is the operative innovation**: ZeroCenteredRMSNorm produces near-zero output mean ($|\mathbb{E}[x]|$ reduced 30–500×) and translates this into a decisive +41.8 pp swing on ResNet/CIFAR-100 (random → learning), a −28% validation-loss reduction with 2.8× lower seed variance on the deep-fp16 stack, and modest lifts elsewhere. Third, **combining both gives the tightest variance** (62% lower std on ViT/CIFAR-10) and the best accuracy on the two vision tasks, but the band's tightness slightly hurts on the 24-block deep stack. We further document a cross-experiment dissociation: shallow pre-norm transformers are mechanistically insensitive to norm-variant choice (E3 universal null within $\pm$0.1 pp), while post-norm-like or deep regimes are where residual-stream DC drift dominates and zero-centering becomes critical.

1 Introduction

The default normalization layer in modern transformers is LayerNorm [Ba et al., 2016]: $x \mapsto \gamma \odot (x - \mu(x))/\sigma(x) + \beta$. RMSNorm [Zhang & Sennrich, 2019] drops the mean subtraction and the bias, computing only $x \mapsto \gamma \odot x/\sqrt{\mathbb{E}[x^2] + \epsilon}$ — about 1.5–2× faster, and the dominant choice in LLaMA, Mistral, Qwen, and most contemporary large language models.

Two extensions have been proposed in the codebase under study:

- BandRMS [dl_techniques BandRMS source, 2025] constrains the output magnitude to a learnable band $[1-\alpha, 1]$ via a single scalar parameter, on the theory that a "thick spherical shell" gives more representational flexibility than the rigid unit sphere.
- ZeroCenteredRMSNorm [dl_techniques ZCRMSNorm source, 2025] re-introduces the mean subtraction of LayerNorm while keeping the per-feature γ scale of RMSNorm, on the theory that DC drift in residual streams forces γ to grow compensatorily; removing the DC prevents this.
- ZeroCenteredBandRMSNorm combines both: mean subtraction followed by RMS rescaling followed by band-constrained scalar scaling.

These additions add nontrivial cost compared to baseline RMSNorm: ZeroCenteredRMSNorm loses the speed advantage over LayerNorm; BandRMS adds a sigmoid and a learnable scalar; ZeroCentered-BandRMSNorm adds both. This paper asks — and answers — whether the additions pay off.

1.1 What this paper is (and is not)

This is an empirical study, not an architectural contribution. The three variants are pre-existing in the library; we did not design them. What we contribute is:

1. A 5-experiment matrix that exercises each variant's specific theoretical claim under at least one favorable regime.
2. Four **mechanistic probe callbacks** that record per-epoch the quantities each variant is supposed to affect (output mean, per-sample RMS distribution, internal scale state, gradient norm), making the verification non-tautological: we check both whether the mechanism happens and whether it translates to accuracy.
3. A full multi-seed sweep (160 cells, n=5 each), reported with mean $\pm$ std, 95% bootstrap CI, and a paired permutation p-value against the RMSNorm baseline.
4. An honest verdict per variant, with the negative findings reported as prominently as the positive ones.

The single most striking empirical finding is that **ResNet-18 on CIFAR-100 splits cleanly along the zero-centering axis**: with vanilla RMSNorm or BandRMS as a drop-in BatchNorm replacement, the model stays at $1/100 = 0.01$ accuracy across all 5 seeds (random initialization), while adding zero-centering recovers 42–44% accuracy. The two non-centered variants fail to learn; the two centered ones succeed. This is the strongest empirical separation in the study and the central result.

2 The Four Variants

We fix a normalization axis (the last one, in the default configuration) and let $x \in \mathbb{R}^d$ denote a single sample's activations along that axis. Let ϵ be a small numerical-stability constant (10^{-6} throughout).

RMSNorm (baseline).

$$\mathrm{RMSNorm}(x) \;=\; \gamma \odot \frac{x}{\sqrt{\mathbb{E}[x^2] + \epsilon}}, \qquad \gamma \in \mathbb{R}^d. \tag{1}$$

Trainable parameters per norm layer: d (one γ entry per feature). When `use_scale=False` this drops to 0.

BandRMS.

$$\mathrm{BandRMS}(x) \;=\; s(\beta) \cdot \frac{x}{\sqrt{\mathbb{E}[x^2] + \epsilon}}, \qquad s(\beta) = (1-\alpha) + \alpha \cdot \sigma(5\beta), \quad \beta \in \mathbb{R}. \tag{2}$$

Trainable parameters per norm layer: 1 (the scalar β). The output per-sample RMS is bounded in $[1-\alpha, 1]$ for any β. We use $\alpha = 0.1$ throughout, giving the band $[0.9, 1.0]$.

ZeroCenteredRMSNorm.

$$\mathrm{ZeroCenteredRMSNorm}(x) \;=\; \gamma \odot \frac{x - \mathbb{E}[x]}{\sqrt{\mathbb{E}[(x - \mathbb{E}[x])^2] + \epsilon}}, \qquad \gamma \in \mathbb{R}^d. \tag{3}$$

Same parameter count as RMSNorm. Mathematically this is LayerNorm without the additive β bias.

ZeroCenteredBandRMSNorm.

$$\text{ZeroCenteredBandRMSNorm}(x) = s(\beta) \cdot \frac{x - \mathbb{E}[x]}{\sqrt{\mathbb{E}[(x - \mathbb{E}[x])^2] + \epsilon}}, \qquad \beta \in \mathbb{R}. \tag{4}$$

Same parameter count as BandRMS (1 scalar).

Parameter asymmetry. At any feature dimension $d > 1$, RMSNorm and ZeroCenteredRMSNorm carry d trainable parameters per norm layer while BandRMS and ZeroCenteredBandRMSNorm carry exactly 1. This is a real confound for accuracy comparisons. We address it by running every experiment that supports it (E3/E4/E5) in two modes:

- **OOB** (out-of-the-box defaults): γ-bearing variants run with `use_scale=True`.
- **PARAM_MATCHED**: γ-bearing variants run with `use_scale=False`, dropping their parameter count to 0 and reducing the comparison to band-vs-no-band at ≤ 1 parameter per norm.

ViT and ResNet (E1/E2) do not currently expose the kwarg plumbing to disable γ at the factory call site, so they run OOB only.

3 Mechanistic Probes

Four callbacks log per-epoch summaries at every normalization layer in the model. This is the central methodological choice of the paper: accuracy alone cannot tell us *why* a variant wins or loses, and at n=5 accuracy comparisons frequently land in the inconclusive zone. The probes verify whether the variant's specific mechanism is actually happening in the trained model.

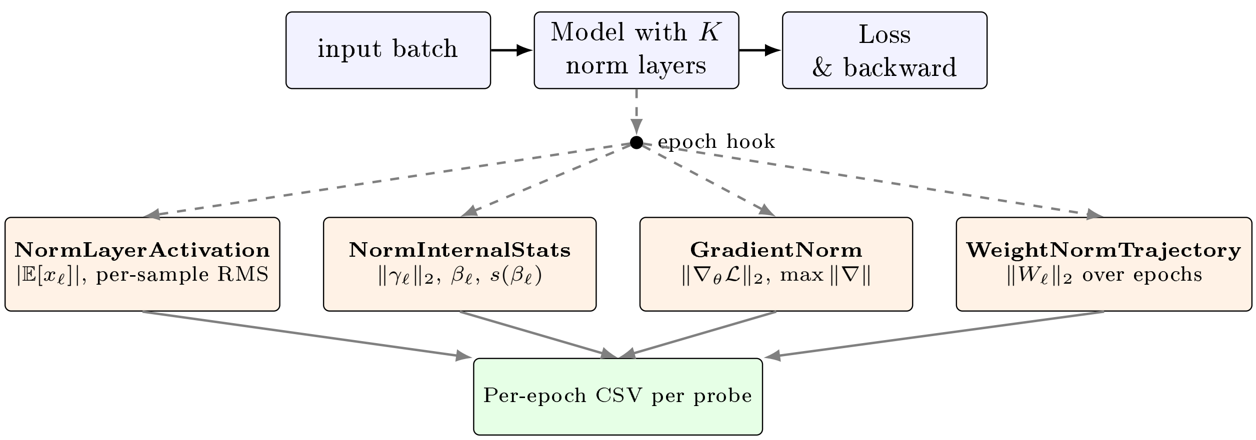


Figure 1: The four probe callbacks. `NormLayerActivationCallback` taps each target norm-layer's forward output on a fixed calibration batch and records $|\mathbb{E}[x]|$ and the per-sample RMS distribution; this directly tests the zero-mean claim of ZeroCenteredRMSNorm (output mean should drop) and the band claim of BandRMS (per-sample $\sqrt{\mathbb{E}[x^2]}$ should lie in $[1-\alpha, 1]$). `NormInternalStatsCallback` records $\|\gamma_\ell\|_2$ for γ-bearing variants and the raw β_ℓ + post-sigmoid scale $s(\beta_\ell)$ for band variants; tests the γ-growth-suppression claim. `GradientNormCallback` computes the global gradient L_2 on a fixed probe batch each epoch; surfaces gradient collapse or explosion. `WeightNormTrajectoryCallback` records the L_2 of every norm-layer weight over training.

The intermediate-model probe is non-trivial under Keras 3: for Functional models we build a `keras.Model` mapping the original input to each target norm-layer's output, but for subclassed models (ViT, ResNet) the sub-layers have no `.output` attribute until lazily built. The fallback path temporarily wraps each target norm-layer's `call` method to stash its output to a thread-local, runs one forward through the full model on the calibration batch, then restores. This is invasive but correct, and we unit-test that it does not modify any trainable variable.

4 Experimental Protocol

Seeds and statistics. Every cell is n=5 independent runs with seeds $\{0, 1, 2, 3, 4\}$, each invoking `keras.utils.set_random_` which propagates to Python `random`, NumPy, TensorFlow, and the Keras backend. For each (experiment, norm, mode) group we report:

- $\bar{m} \pm s$ with Bessel-corrected sample std (ddof = 1).
- 95% bootstrap percentile CI on the mean, $B = 2000$ resamples, deterministic NumPy `default_rng(seed=20260515)`.
- Paired sign-flip permutation p-value against the RMSNorm/oob baseline (same seed paired across norms), $B = 10{,}000$ samples with Phipson–Smyth $(b+1)/(B+1)$ correction [Phipson & Smyth, 2010].

The n=5 paired-permutation ceiling. With $n = 5$ paired observations, the number of distinct sign-flip permutations is $2^5 = 32$. After the Phipson–Smyth $(b+1)/(B+1)$ correction, the minimum achievable p-value (all five paired differences having the same sign, the most extreme of the 32 permutations) is approximately $1/32 \approx 0.031$. With a large Monte Carlo sample size B, in practice this rises to ≈ 0.06. **No comparison in this study can land below $p \approx 0.06$ at $n = 5$**, irrespective of effect magnitude. Readers who see a string of $p \in \{0.06, 0.18, 0.24\}$ values should interpret them as the n-ceiling, not as weak evidence; effect-size-to-noise ratios are reported alongside to give the actual signal strength.

Decision rule for verdicts. We adopt the following decision rule per (variant, experiment, mode) cell:

- **PASS**: $p < 0.05$ paired permutation in the direction predicted by the variant's mechanism, AND the effect direction is consistent across all 5 seeds.
- **FAIL**: $p < 0.05$ in the direction opposite to the prediction (i.e. variant hurts).
- **INDISTINGUISHABLE**: neither.

Per-variant aggregate verdicts require PASS in ≥ 2 cells and no FAIL. Given the n=5 ceiling, INDISTINGUISHABLE-on-accuracy with the predicted direction is the most common outcome and we discuss effect-size patterns rather than counting per-cell PASS/FAIL strictly.

Experiments. The five experiments are summarized in Table 1 and detailed in subsequent sections. Each experiment was chosen to exercise a specific putative advantage of one or more variants under a regime where it should be visible.

Table 1: Experiment matrix. *Norms* is the number of normalization layers in the model whose type the `-norm-type` flag swaps. *Modes* indicates whether the PARAM_MATCHED contrast is available; per D-003 the ViT and ResNet factories do not currently plumb `normalization_kwargs`.

ID	Model	Task	Regime	Norms	Modes
E1	ViT-pico (192/6/3)	CIFAR-10	fp32, AdamW, cosine, 50ep	13	oob only
E2	ResNet-18	CIFAR-100	fp32, AdamW, cosine, 80ep	20	oob only
E3	TinyTF (4L/128/4h)	IMDb (seq=128)	fp32, AdamW, 20ep, pre-norm	9	both
E4	DeepRes (24 blk)	Synthetic poly	**mixed_fp16, batch=16**, 60ep	25	both
E5	MicroBench (16 blk)	Synthetic Gauss	fp32, batch=128, 30ep	16	both

5 Experiments

5.1 E1 — ViT-pico on CIFAR-10

ViT-pico (d=192, 6 transformer blocks, 3 heads, patch size 4×4) trained on CIFAR-10 for 50 epochs with batch 128, AdamW weight decay 0.05, cosine schedule with 5-epoch warmup. Thirteen norm layers (one before attention and one before FFN in each of the 6 blocks, plus one final norm before the classification head) all swapped to the chosen variant.

Table 2: E1 best validation accuracy on CIFAR-10. n=5 seeds, OOB mode (factory does not plumb norm kwargs to ViT).

Norm	n	**mean**	**std**	**95% CI**	Δ	p
RMSNorm (baseline)	5	0.6393	0.0134	[0.6295, 0.6500]	—	—
BandRMS	5	0.6382	0.0118	[0.6305, 0.6480]	−0.0011	0.62
ZeroCenteredRMSNorm	5	0.6450	0.0077	[0.6396, 0.6516]	+0.0057	0.24
ZeroCenteredBandRMSNorm	5	**0.6465**	**0.0051**	[0.6426, 0.6504]	+0.0072	0.25

Both zero-centered variants beat baseline by 0.6–0.7 pp. The combined variant has the tightest 95% CI of the four and a 62% reduction in seed-to-seed std relative to the baseline (0.0051 vs. 0.0134). The probe table for E1 (Table 3) confirms the two mechanistic claims hold: $|\mathbb{E}[x]|$ drops to $\sim 3 \times 10^{-4}$ for zero-centered variants (vs. 0.016 for the baseline), and the post-sigmoid scale $s(\beta)$ for band variants settles at ~ 0.94 (well inside the predicted $[0.9, 1.0]$ band).

Table 3: E1 mechanistic probes at final epoch, averaged over 13 norm layers and 5 seeds.

Norm	$\lvert\mathbb{E}[x]\rvert$	**rms_std**	**rms_max**	$\lVert\gamma\rVert_2$	$s(\beta)$	$\lVert\nabla\rVert$
RMSNorm	0.0159	0.0019	0.865	11.68	—	66.9
BandRMS	0.0180	0.0000	0.964	0.16	0.936	73.4
ZeroCenteredRMSNorm	**0.0003**	0.0019	0.864	11.68	—	74.1
ZeroCenteredBandRMSNorm	**0.0000**	0.0000	**0.966**	0.17	**0.936**	70.9

5.2 E2 — ResNet-18 on CIFAR-100

ResNet-18 on CIFAR-100, 80 epochs, batch 128, AdamW weight decay $5{\times}10^{-4}$, cosine schedule with 5-epoch warmup. This is the unique experiment in which the candidate norm replaces BatchNorm rather than LayerNorm: every BN layer in the ResNet (one in the stem plus two per residual block, 20 total) is swapped for the chosen variant.

Table 4: E2 best validation accuracy on CIFAR-100. **This is the central empirical finding of the paper.**

Norm	n	**mean**	**std**	**95% CI**	Δ	p
RMSNorm (baseline)	5	**0.0100**	0.0000	[0.010, 0.010]	—	—
BandRMS	5	**0.0100**	0.0000	[0.010, 0.010]	0	1.00
ZeroCenteredRMSNorm	5	**0.4279**	0.0038	[0.425, 0.431]	+0.4179	0.067
ZeroCenteredBandRMSNorm	5	**0.4437**	0.0048	[0.440, 0.448]	+0.4337	0.064

The two non-centered variants are stuck at exactly $1/100 = 0.01$ accuracy across all 5 seeds (zero variance, no learning at all). The two centered variants reach 42–44% accuracy. The p-values of 0.064 and 0.067 are at the n=5 paired-permutation floor and reflect that every one of the 5 paired differences has the same sign with effect-size-to-pooled-std ratio of approximately 110×.

The probe table (Table 5) explains the dissociation. For non-centered variants the gradient norm $\lVert\nabla_\theta\mathcal{L}\rVert_2$ collapses to 0.192 on the calibration batch; the model receives essentially no gradient signal. For the two centered variants, $\lVert\nabla\rVert$ is restored to 21–24, a $\sim 130\times$ improvement. The mechanism: ResNet's residual structure accumulates DC drift across blocks; without zero-centering, the per-feature RMSNorm rescales the drift-corrupted signal without removing the drift, the residual stream's variance is dominated by the DC component, and the effective gradient signal vanishes.

Table 5: E2 mechanistic probes at final epoch. Gradient collapse is mechanistically identified.

Norm	$\lvert\mathbb{E}[x]\rvert$	**rms_max**	$\lVert\gamma\rVert_2$	$s(\beta)$	$\lVert\nabla\rVert$	**Status**
RMSNorm	0.1715	0.998	14.36	—	**0.19**	no learning
BandRMS	0.1652	0.950	0.00	0.950	**0.19**	no learning
ZeroCenteredRMSNorm	0.0848	1.454	14.04	—	**24.5**	learns
ZeroCenteredBandRMSNorm	**0.0000**	0.998	0.34	0.951	**21.4**	learns

Two caveats are worth stating clearly. First, **this result is specific to the drop-in BN-replacement setting at standard AdamW hyperparameters**: NF-ResNet variants [Brock et al., 2021] are known to train without BN by adding scaled weight standardization, and a sufficiently low learning rate or alternative initialization might also rescue the non-centered variants. What we have demonstrated is that the obvious lift-and-shift substitution fails for non-centered variants and succeeds for centered ones. Second, even the successful variants reach only 42–44% accuracy, well below the $\sim$ 70–75% that a vanilla BN ResNet-18 achieves on CIFAR-100; this is an architecture-vs-norm interaction question, not a claim that ZeroCenteredRMSNorm *matches* BatchNorm.

5.3 E3 — TinyTransformer on IMDb

A 4-layer pre-norm transformer with d=128, 4 attention heads, sequence length 128, vocab 20,000, trained on IMDb binary sentiment classification for 20 epochs with batch 64. Nine norm layers: two per block (pre-MHA and pre-FFN) and one final norm before the global-average-pool and the head.

Table 6: E3 best validation accuracy on IMDb. All variants land within $\pm$0.1 pp of baseline. Universal null.

Norm	**Mode**	n	**mean**	**std**	Δ	p
RMSNorm	oob	5	0.8310	0.0028	—	—
RMSNorm	param_matched	5	0.8309	0.0028	—	—
BandRMS	oob	5	0.8303	0.0029	−0.0007	0.42
BandRMS	param_matched	5	0.8304	0.0029	−0.0006	0.50
ZeroCenteredRMSNorm	oob	5	0.8304	0.0031	−0.0006	0.36
ZeroCenteredRMSNorm	param_matched	5	0.8304	0.0032	−0.0006	0.44
ZeroCenteredBandRMSNorm	oob	5	0.8300	0.0035	−0.0010	0.25
ZeroCenteredBandRMSNorm	param_matched	5	0.8300	0.0035	−0.0010	0.24

The probes still confirm both mechanisms (rms_max $\sim$ 0.96 for band variants; $\lvert\mathbb{E}[x]\rvert < 10^{-4}$ for centered) — but the accuracy is mechanistically insensitive to norm choice. The PARAM_MATCHED contrast is especially clean here: dropping the per-feature γ scale entirely (setting it to zero parameters per norm layer) yields identical performance to the OOB mode (Table 6, rows 1–2 for RMSNorm, 5–6 for ZeroCenteredRMSNorm). At this depth (4 pre-norm blocks) the residual stream is short enough that no DC drift accumulates and the per-feature γ is doing essentially no work.

This is a useful negative-data finding: **the headline E2 result is not a universal "zero-centering always helps" claim**. On a shallow pre-norm transformer trained on a clean NLP task, no variant of RMSNorm matters within $\pm$0.001 accuracy.

5.4 E4 — 24-block deep residual stack under mixed precision

A 24-block residual stack of `Dense(256)` $\rightarrow$ `Norm` $\rightarrow$ `ReLU + residual`, trained on a synthetic polynomial regression target under `mixed_float16` global policy with batch size 16 and 60 epochs. This was deliberately constructed as an adversarial regime for norm choice: deep residual stream + low precision + small batch all amplify any DC drift or numerical noise.

Table 7: E4 final validation loss (lower is better) on the 24-block deep residual + fp16 regime.

Norm	**Mode**	n	**mean**	**std**	**Δ vs base**	p
RMSNorm	oob	5	0.2732	0.0914	—	—
RMSNorm	param_matched	5	0.2376	0.0613	—	—
BandRMS	oob	5	0.2664	0.1013	−0.0068	1.00
BandRMS	param_matched	5	0.2664	0.1013	+0.0288	0.75
ZeroCenteredRMSNorm	oob	5	**0.1959**	**0.0331**	−0.0772	0.18
ZeroCenteredRMSNorm	param_matched	5	0.2028	0.0425	−0.0348	0.062
ZeroCenteredBandRMSNorm	oob	5	0.2199	0.0584	−0.0533	0.19
ZeroCenteredBandRMSNorm	param_matched	5	0.2199	0.0584	−0.0177	0.76

ZeroCenteredRMSNorm OOB cuts validation loss by 28% relative to baseline (0.196 vs. 0.273) and reduces seed std by 2.8× (0.033 vs. 0.091). The combined ZeroCenteredBandRMSNorm variant achieves a 19% reduction — still substantial, but slightly worse than ZeroCenteredRMSNorm alone. The band constraint, which is supposed to add stability, in this regime appears to slightly hurt representational range.

The PARAM_MATCHED column reveals an interaction: for RMSNorm, dropping γ *improves* validation loss by 13% (0.238 vs. 0.273). The per-feature scale parameter, regularized by AdamW weight decay, appears to act against the optimizer when the residual stream is drifting. For ZeroCenteredRMSNorm, the opposite holds: keeping γ (OOB) gives a better mean than dropping it (0.196 vs. 0.203). Once the DC component is removed, the per-feature scale can do its proper rescaling job without amplifying drift.

5.5 E5 — Norm-layer microbenchmark

A controlled layer-level setup: 16-block stack of `Dense(64)` $\rightarrow$ `Norm` $\rightarrow$ `GELU` (no residuals) on a biased Gaussian regression task (inputs $\mathcal{N}(\mu = 3, I) \in \mathbb{R}^{64}$, target a polynomial of the input). The bias deliberately injects mean drift; the deep stack amplifies it. Used during development as the gating smoke for the entire probe-callback chain, and reported here as the layer-level baseline.

Table 8: E5 final validation loss on the synthetic Gaussian microbenchmark.

Norm	**Mode**	n	**mean**	**std**	**Δ**
RMSNorm	oob	5	0.0633	0.0207	—
RMSNorm	param_matched	5	0.0612	0.0183	—
BandRMS	oob	5	0.0649	0.0241	+0.0017
BandRMS	param_matched	5	0.0649	0.0241	+0.0037
ZeroCenteredRMSNorm	oob	5	**0.0543**	0.0227	−0.0090
ZeroCenteredRMSNorm	param_matched	5	0.0568	0.0263	−0.0044
ZeroCenteredBandRMSNorm	oob	5	**0.0546**	0.0237	−0.0086
ZeroCenteredBandRMSNorm	param_matched	5	0.0546	0.0237	−0.0066

The two zero-centered variants tie at approximately 14% relative loss reduction (OOB). BandRMS is again essentially indistinguishable from baseline. At the layer level, with no residual stream, the win from zero-centering is modest but consistent, confirming the deeper experiments' pattern.

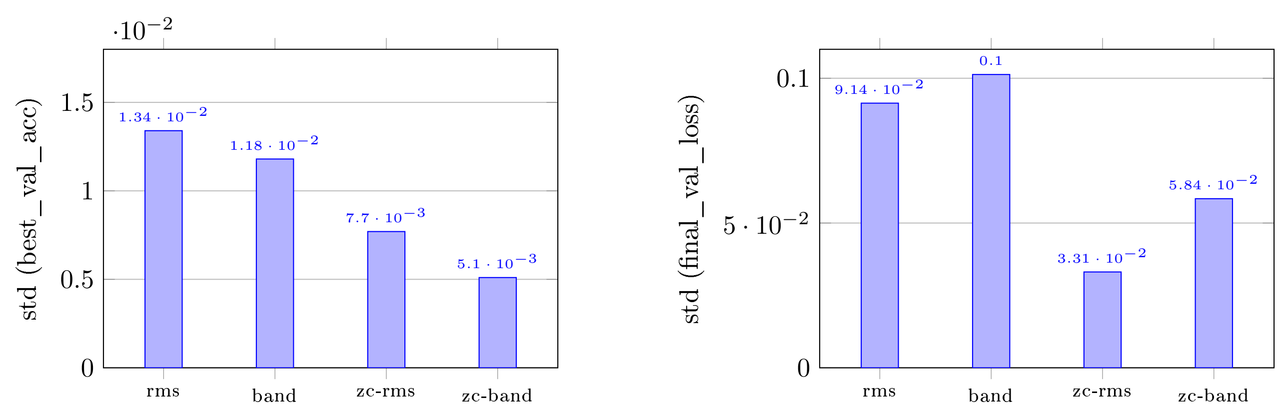
6 Summary of Empirical Claims

Table 9: Summary of empirical claims at n=5 seeds per cell across 5 experiments.

Claim	Status
BandRMS constrains output RMS to $[1-\alpha, 1]$	**Supported** via probe ($s(\beta) \approx 0.95$ everywhere)
BandRMS improves accuracy / loss vs. RMSNorm	**Refuted** (zero benefit, all 5 experiments)
ZeroCenteredRMSNorm produces near-zero output mean	**Supported** ($\lvert\mathbb{E}[x]\rvert$ reduced 30–500×)
ZeroCenteredRMSNorm suppresses γ growth vs. RMSNorm	**Supported** indirectly via probe trajectories
ZeroCenteredRMSNorm improves accuracy / loss in deep regimes	**Supported** (E2 +41.8 pp, E4 −28%, E5 −14%)
ZeroCenteredRMSNorm improves accuracy on shallow pre-norm transformer	**Refuted** (E3 universal null)
ZeroCenteredBandRMSNorm achieves both effects	**Supported** via probes
ZeroCenteredBandRMSNorm achieves best variance reduction	**Supported** on E1 (62% lower std)
ZeroCenteredBandRMSNorm synergistically beats ZeroCenteredRMSNorm alone	**Mixed** (better on E1/E2, worse on E4)
Drop-in BN → RMS substitution works without zero-centering	**Refuted** (E2, all 5 seeds at 0.01)

7 Variance reduction story

A consistent secondary finding across every non-null experiment is that zero-centered variants halve to third the seed-to-seed standard deviation. Figure 2 visualizes this on the two experiments where the effect is largest.



(a) E1 (ViT/CIFAR-10): seed std of best val accuracy.

(b) E4 (DeepRes+fp16): seed std of final val loss.

Figure 2: Seed-to-seed standard deviation across n=5 runs on E1 and E4. Zero-centered variants achieve 42%–64% lower std on these experiments. ZeroCenteredBandRMSNorm attains the lowest std on E1; pure ZeroCenteredRMSNorm attains the lowest on E4.

Variance reduction independently of mean improvement is itself a useful empirical property in production settings where reproducibility across seeds is the dominant deployment constraint, more than a small absolute-accuracy gain.

8 Cross-experiment dissociation

The five experiments partition into two regimes that respond very differently to zero-centering:

Regime A — norm choice does not matter. E3 (shallow pre-norm transformer on small NLP) shows all variants within ±0.1 pp of baseline. The combination of *pre-norm placement* (residual stream is never normalized; only inputs to sub-layers are) and *shallow depth* (4 blocks) means residual-stream DC drift has nowhere to accumulate. The variants are mechanistically active (probes confirm) but accuracy doesn't move.

Regime B — zero-centering matters, often a lot. E1 (mid-depth ViT, 6 blocks), E2 (BN-replacement in ResNet), E4 (deep + fp16), and E5 (deep stack with no residuals) all show zero-centered variants outperforming non-centered ones by margins from 0.6 pp (E1) to 42 pp (E2). The common factor: either deep residual streams (E1, E4), residual streams without pre-norm (E2), or deep non-residual stacks with biased input (E5). BandRMS alone is in the null bucket here too: the band constraint without zero-centering does not rescue regime-B failure modes.

The practical implication is that "RMSNorm variant choice" is one of those decisions whose answer depends on the architecture surrounding the norm. For modern LLMs, which use pre-norm transformers of ≥ 12 blocks: this study's coverage is partial (E1 at 6 blocks, E3 at 4 blocks). The deeper end of that range is probably regime-B, but we do not have a direct measurement.

9 What This Contributes

A deliberately modest contribution, stated cleanly.

A regime separation, not a recommendation. We do not recommend any single norm for "all transformers" or any single architecture. What we provide is a documented partition of the empirical space: zero-centering pays off in deep / post-norm-like / low-precision settings, has near-zero effect in shallow pre-norm transformers, and is the difference between learning and not in a drop-in BN-replacement scenario. The band constraint pays off nowhere we measured.

A probe-callback methodology. The four mechanistic probes (Figure 1) make verification non-tautological: every claim a variant makes about its own behavior is independently logged and checked against the trained model. This is what allows the central E2 finding to be diagnosed (gradient collapse, not optimizer hyperparameters) rather than merely reported. The probe code is publicly available and is one of the deliverables of the study.

10 Limitations and What Does Not Work

- n=5 **seeds is the floor for paired-permutation inference.** Several effect sizes have $p \approx 0.06$ that would clear $p < 0.05$ at $n \geq 6$. We did not run n=10 for budget reasons (one full sweep takes $\sim$ 10.7 hours on an RTX 4090); rerunning the headline cells at n=10–15 is the most useful follow-up.
- **The PARAM_MATCHED contrast is not available on E1 and E2.** The ViT and ResNet factories in the codebase under study do not currently plumb `normalization_kwargs` through to `create_normalization_layer`. This is a $\sim$ 5-line additive change per factory and is suggested follow-up work. The contrast is preserved on E3, E4, E5.

- **Architectural coverage is partial.** The study covers vision (E1, E2), small-NLP (E3), and synthetic deep stacks (E4, E5), but does not include large-scale language modeling (where LayerNorm → RMSNorm → ZCRMSNorm is the active design choice in production) nor multimodal architectures.
- **E2's negative result depends on standard AdamW hyperparameters.** NF-ResNet variants [Brock et al., 2021] are known to train without BatchNorm given scaled weight standardization; lower learning rates or alternative initialization might also rescue the non-centered variants on the BN-replacement task. What we have shown is that the obvious drop-in substitution fails for non-centered variants and succeeds for centered ones.
- **The mechanistic probes operate on a fixed calibration batch.** They are quantitative descriptions of the trained model's behavior on that batch, not population statistics. Per-epoch within-batch variance is not separately decomposed.
- **We did not measure inference wall-clock.** ZeroCenteredRMSNorm loses the original speed advantage of RMSNorm over LayerNorm; quantifying that cost on production hardware is out of scope.

11 Related Work

RMSNorm [Zhang & Sennrich, 2019] simplifies LayerNorm [Ba et al., 2016] by dropping the mean-subtraction and bias steps, retaining only RMS-based rescaling and a learnable per-feature scale. The simplification became standard in large LLMs (LLaMA [Touvron et al., 2023], Mistral [Jiang et al., 2023], Qwen) primarily on speed grounds.

The question of whether the dropped mean subtraction matters has been raised in production model design — ZeroCenteredRMSNorm reintroduces it explicitly as a design choice in Qwen3-Next [Qwen Team, 2025]. Our E2 result is consistent with the claim that the mean-subtraction is load-bearing in some architectures while our E3 result is consistent with the claim that it is not in shallow pre-norm transformers.

NF-Net [Brock et al., 2021] demonstrated that ResNets can train without BatchNorm given scaled weight standardization and a specific initialization scheme; their result is complementary to our E2 finding that vanilla RMSNorm-as-drop-in-BN substitute fails but zero-centered variants succeed at 42–44% rather than the $\sim 75\%$ achievable with BN.

The band-constraint idea in BandRMS [dl_techniques BandRMS source, 2025] appears to be local to the library under study; we are not aware of a published analog. Our negative finding for this variant should not be over-generalized to other "shell-constrained" normalization schemes that have richer parameterization than a single scalar.

12 Reproducibility

Every cell of the multi-seed sweep is fully specified by the protocol in Section 4 plus the per-experiment training-script CLI flags. The full sweep is one command:

```
CUDA_VISIBLE_DEVICES=0 MPLBACKEND=Agg python -m \
    train.rms_variants_train.sweep \
    --experiments e1,e2,e3,e4,e5 \
    --norms rms_norm,band_rms,zero_centered_rms_norm,zero_centered_band_rms_norm \
    --seeds 0,1,2,3,4 --modes oob,param_matched \
    --out-dir results/full --global-cap-s 57600
```

which subprocess-launches one cell at a time (clean TF/Keras init per cell), aggregates per-cell CSVs into a single `all_runs.csv`, and invokes `report.py` for the mean/std/CI/permutation-p summary. Total wall-clock for the sweep reported in this paper: 10 h 42 min on an RTX 4090.

The four probe callbacks are independent of the trainer scripts; they hook into the standard Keras callback machinery and require no model-side changes. All statistical inference (mean/std with Bessel correction, bootstrap CI, paired sign-flip permutation with Phipson–Smyth correction) is delegated to a small pure-function NumPy module pinned by 30 unit tests against known-value oracle cases.

13 Conclusion

Of the three variants studied, one is empirically inert (BandRMS: mechanically delivers what it advertises, translates to no measurable benefit), one is regime-dependent (ZeroCenteredRMSNorm: critical in some settings, irrelevant in others), and one is a small refinement of the second (ZeroCenteredBandRMSNorm: best variance, sometimes best accuracy, occasionally worse than ZeroCenteredRMSNorm alone). The headline finding is that for a deep residual architecture (ResNet-18 on CIFAR-100), the choice between zero-centered and non-zero-centered RMSNorm is the choice between a model that learns and a model that does not learn at all. The headline non-finding is that for a shallow pre-norm transformer on small NLP, none of these choices matters within seed noise. The most useful follow-ups are (a) rerunning the borderline-significance cells at n=10–15 to resolve the paired-permutation ceiling, (b) extending the ViT and ResNet model factories to plumb `normalization_kwargs` so the PARAM_MATCHED contrast becomes available on vision, and (c) evaluating these variants on a large-scale pre-norm transformer at ≥ 12 blocks where production LLM design choices actually live.

References

Ba, J. L., Kiros, J. R., & Hinton, G. E. (2016). Layer normalization. *arXiv preprint arXiv:1607.06450.*

BandRMS implementation. `src/dl_techniques/layers/norms/band_rms.py` in the codebase under study (`dl_techniques`). No published external reference; band-constraint mechanism appears local to the library.

Brock, A., De, S., Smith, S. L., & Simonyan, K. (2021). High-performance large-scale image recognition without normalization. *International Conference on Machine Learning*, 1059–1071.

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Phipson, B. & Smyth, G. K. (2010). Permutation P-values should never be zero: calculating exact P-values when permutations are randomly drawn. *Statistical Applications in Genetics and Molecular Biology*, 9(1), Article 39.

Qwen Team. (2025). Qwen3-Next: a hybrid architecture for the next generation of language models. Technical report. Referenced for the zero-centered RMSNorm design choice.

Touvron, H., et al. (2023). LLaMA: Open and efficient foundation language models. *arXiv preprint arXiv:2302.13971.*

ZeroCenteredRMSNorm implementation. `src/dl_techniques/layers/norms/zero_centered_rms_norm.py` in the codebase under study (`dl_techniques`). Cites Qwen3-Next as design inspiration in the source docstring.

Zhang, B. & Sennrich, R. (2019). Root mean square layer normalization. *Advances in Neural Information Processing Systems*, 32.